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Editors
Physical Review Letters

Dear Editors,

Please consider our manuscript, “Winding-Free Exact Diagonalization of Quantum Spin Ice,” for publication in *Physical Review Letters*.

The recent two-peak heat capacity of $Ce_2Hf_2O_7$ creates a concrete need to connect high-temperature exchange fits to the low-energy coherent ice dynamics. In the relevant sign-problem regime, exact diagonalization is one of the few unbiased dynamical tools, but its standard 32-site FCC torus supports a four-spin exchange that closes only through a periodic boundary. Because this artifact enters at second order while the physical pyrochlore-hexagon ring exchange enters at third order, it dominates precisely where the quantum-spin-ice expansion is best controlled.

We derive a gauge-invariant transported-dipole label for complete virtual processes and show that its zero-character projection becomes a simple polarization-parity mask on the downfolded ice Hamiltonian. On the complete 2,970-state FCC-32 ice manifold, explicit path filtering and the mask agree entrywise through third order. The protocol removes all wrapping four- and six-site processes, retains every contractible hexagon with unchanged amplitude, restores cubic scaling with the contractible-ring energy, and reorganizes the four-sublattice-traced longitudinal pseudospin structure factor at all eight allowed cluster momenta.

For $Ce_2Hf_2O_7$, the published seventh-order NLC parameter set places the clean FCC-32 ring peak near 7.1 mK, well below the observed 25 mK feature. We derive a high-temperature-moment-constrained parameter candidate that places the ring peak at 25 mK while preserving the leading exchange second moment and transverse anisotropy of the published set. We clearly identify this candidate as a target for a new NLC evaluation rather than as a completed linked-cluster refit.

We believe the manuscript is appropriate for PRL because it provides an operator-level correction to a parametrically large finite-size error in a sign-problem setting, validates the correction rigorously on FCC-32, and turns a newly observed material scale into a concrete constraint on future joint linked-cluster and winding-free calculations.

This manuscript is original, is not under consideration elsewhere, and has been approved by all authors. [Add related-manuscript and preprint statement, if applicable.]

Sincerely,

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